

Parity and checksum

Links drop bits

Detect, do not correct

- Even parity sets the check bit so the total ones count is even
- Odd parity flips that sense
- XOR of all bits is ones-count mod two
- Parity does not correct bit errors
- A multi-byte XOR checksum verifies when the fold comes back zero

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Parity / XOR checksum

Compute **even/odd parity**, see **reduction XOR**, and build a simple multi-byte XOR checksum — then flip a bit and watch it fail.

Starter example: Data 0x2A (three 1s). Compute even parity, attach it, verify, then flip a data bit — check fails.

Load starter example

Challenges 0 / 22 cleared

Quiz: reduce: XOR of all bits equals ones mod 2 Answer: 2

Answer

Show hintCheckNextIdle

Quiz: reduceQuiz: even PQuiz: detectQuiz: HDLStarter 0x2AEven parityOdd parityAttach & verify

Flip failsXOR checksumChecksum OKCorrupt checksumQuiz: all zeroQuiz: 0xFF evenToggle ones

Quiz: two flipsPreset 0x2AQuiz: csSaw reduceOdd verifyQuiz: crypto?Full flow

Core ideas

Reduction XOR

$\wedge$ of all bits = ones mod 2 — the odd-parity of the word.

Even / odd

Parity bit chosen so total ones (data+parity) is even or odd.

Parity checksum starter

Workbook practice

- In the workbook track, take data hex two-A
- Count the ones, compute even and odd parity bits by hand
- Sketch a three-byte XOR checksum and note that a corrupt byte fails the zero fold
- Name one pitfall: assuming parity repairs the bad bit

Pitfalls to watch

- Do not treat parity as error correction
- Even and odd modes disagree if you mix them across sender and receiver
- And remember: the browser lab is literacy
- Real links still need framing

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, compute one parity bit and one XOR checksum by hand
- When you are ready, take the short quiz, then continue to fixed-point